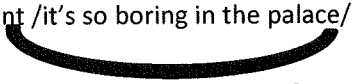
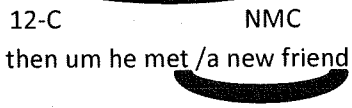
Coding Recalls (evolving draft -- as of 6/12/2012)

Resources needed: causal connection story unit guide; transcribed recall; glossary of coding examples.

1. Parse recall protocol into thought units that match (as possible) story units
    a. Use the parsed story units to guide parsing of the recall. Match to a story unit whenever possible
    b. If words inside a longer thought unit match a separate story unit, draw an arc under the inside words. The arc should connect the two parts of the longer thought unit.
        i. Example: / I want /it's so boring in the palace/ to play inside./





        ii. Example: /And then um he met /a new friend /named Ben/

2. If "(inaudible)" makes it impossible to code, mark ☐ in the margin. Listen to the original recording and insert correct word(s), if audible. Correct the electronic transcript of the recall. Code accordingly. Cross out ☐ after listening.

3. Decide if the parsed recall thought unit directly matches one (or two adjacent) story unit(s), and what code applies.
    a. If it matches,
        i. Write the story unit number(s) above the thought unit.
        ii. Write C, L or (rarely) MI, according to definitions below and story glossaries.
    b. If it does not match or spans multiple non-adjacent story units,
        i. Write NMC or NMI above the thought unit, according to definitions below and story glossaries.

4. Code Definitions
    a. **C** (conservative): The subject's thought unit **directly matches** a story unit and is a literal or nearly literal rendering of the original story unit. Note that a high proportion of conservative clauses reflects (near-) verbatim memory for the text, but few inferences.
    b. **L** (liberal): The subject's thought unit **directly matches** a story unit and captures the gist of the story unit (the essence of its meaning). Paraphrasing may be present. Note that a high proportion of liberal clauses reflects the extent to which the reader has established a coherent representation of the text
    c. **MI** (matched but incorrect)**:** The subject's thought unit directly matches a story unit, but is significantly incorrect. MI is reserved for absolutely and importantly contradictory interpretations, not just an erroneous detail of the story. Since MI is matched, it is followed by its story unit #.
    d. **NMC** (no match consistent): The subject's thought unit **cannot be matched directly** with the gist meaning of any one (or two adjacent) story unit, but it is **valid, consistent with (constrained by) the information in the text**, or plausible. Note that no match

consistent clauses reflect the extent to which the reader has elaborated his/her mental representation of the text through inferences.

e. **NMI** (no match inconsistent): The subject's thought unit is **invalid, inconsistent (unconstrained by) the information in the text**. The thought unit should be flat-out incorrect to be NMI. Note that no match inconsistent clauses suggest that the reader has constructed an erroneous, or partly erroneous, representation of the text.

FAQs (and OAQs) for recall coding

- Q **How literal must a C be?** If the subject's thought unit words literally match words/ideas in a story unit, it is a C. The fact that the original story unit has *additional words/ideas* does not negate the Conservative (C) match.
- Q **But when some words or ideas are missing, how do I decide that the missing words or incomplete idea changes the code from C to L?** If the information that was left out is important to the function or meaning of the current story sentence, then the thought unit is coded L. Importance is evaluated in light of preceding story context. If the missing information is unimportant details, code C.
- Q **How closely verbatim should a C be? Should context drive the decision between C and L?** Generally use the literal or nearly literal rule. A deviation from the original story unit is on a continuum from C to L, that is, in some cases it's hard to decide (it just is!). Ask two or three coders for their *first impressions* (no fair arguing your point of view) and use the majority to decide.

- Q **How to code if pronouns are ambiguous or erroneous?** If the antecedent of the pronoun *could* include the actual antecedent, code C (assuming the rest of the thought unit is C). If the antecedent of the thought unit pronoun is *not* the actual antecedent, but it does not violate the story line, code L (assuming the thought unit matches a story unit and matches the gist, etc.).
- Q **What if the subject uses a wrong word?** Apply the C and L standards. If the student misunderstood a detail of the story text, it is unlikely that the retell will be a conservative match. Whether it is an L or an MI depends on how important the error is.
- Q **Where to parse when the subject hems, haws and repeats?** When subject makes a false start, repeats words or phrases or self-corrects, include them all into the same parsed thought unit (as matched to the story unit). If there is more than one complete thought unit and each would be coded differently, code only the final statement, and code the earlier thought unit as (R).

- Q **How do I differentiate NMI from NMC?** Only code as inconsistent (NMI or MI) if a thought unit ***flat out contradicts the story***. If the *function* of the information is consistent with or *plausibly consistent* with the story, it should be coded NMC or L.

- Q **What's the best criterion for choosing the matched story unit?** When choosing which story unit matches a thought unit, think about the function of the content of each. See Glossary for Sam, Liberal story unit #47.

Q **What if an SU is retold multiple times?** Each story unit can only be represented in the recall one time. If a story unit is repeated or a no-match thought unit is repeated, don't code or count it on subsequent iteration(s). Write (R) to indicate "repeated."

5. Code for order of recall
    a. In the right margin of the recall protocol, write in a vertical list from first to last, the story unit numbers that correspond to the C and L thought units.
    b. The list will likely not be in order or include all story unit #s.

6. Score highly connected (HC) thought units
    a. Highly connected story units are (will be!) in boldface on the causal connection story unit guide.
    b. Using the order of recall list of story unit numbers, find the "highly connected" story units matched in the recall. If the thought unit matches two story units, if either story unit is highly connected, code the thought unit as highly connected.
    c. Next to the code (C or L) write HC in a box ☐ to indicate the thought unit matches a highly connected story unit.
        i. Example: ☒ HC C-26
        ii. Example : ☐ L-8

Annie Helps a Dolphin

# of Conn.	Antecedent Connections	Consequent connections	Story units
7		3,7,11,12, 21, 30, 38	1. Annie (recently moved to a small island)
6		3, 7, 12, 21, 30, 38	2. and her father recently moved to a small island.
5	1, 2	4,5,14	3. Annie was new to the island,
3	3	5,7	4. and had not made any friends.
4	3, 4	49,50	5. She hoped to meet a new friend soon. [GOAL STATEMENT]
2		7,17	6. Every day,
10	1, 2, 4, 6	8, 9, 16, 17, 23, 25	7. Annie watched wild dolphins
4	7	9, 23, 25	8. swimming in groups
3	7, 8	17	9. along the shore.
1		11	10. One day,
4	1, 10	12,13	11. Annie and her father went into town.
4	1,2, 11	13	12. They met a fisherman
5	11, 12	14,15, 49	13. and his daughter, Sue.
3	3, 13	15	14. Both girls were shy,
5	13, 14	16, 17, 18	15. but soon they began to talk.
3	7, 15	17	16. about dolphins
10	6,7,9, 15,16	18,19,21,43,50	17. "I love to watch the wild dolphins swimming along the shore,"
4	15, 17	19,20	18. Annie told Sue.
4	17, 18	20, 43	19. "Me, too!"
4	18, 19	21, 24	20. said Sue.
11	1,2, 17, 20	22,23,25,28,33, 34,35	21. "In fact, I saw one swimming all by itself,
1	21		22. just this morning."
7	7, 8, 21	24,25,26,27	23. "Hmm, that's strange,"
4	20,23	25,29	24. said Annie.
8	7, 8, 21, 23,24	27, 28,33	25. "I thought dolphins usually swim in groups."
1	23		26. "Let's see
6	23, 25	28, 29, 30,31	27. if we can learn more
5	21, 25, 27	29, 31	28. about how dolphins are supposed to behave,"
3	24, 27, 28		29. suggested Sue.
4	1, 2, 27	31	30. They went to the small library in town,
5	27, 28, 30	32,33	31. and looked for some books about dolphins.
2	31	36	32. They learned

Annie Helps a Dolphin

5	21, 25, 31	34, 36	33. that dolphins normally do not swim alone.
7	21, 33	36, 37,38,39,41	34. "Maybe the dolphin was lost
1	21		35. you saw
3	32, 33, 34		36. said Annie.
2	34	38	37. They decided
7	1, 2, 34, 37	39,40,42	38. to call the animal rescue office on the island
3	34, 38	40	39. to ask for help.
3	38, 39	41	40. Sure enough, the rescuers discovered
4	34, 40	42, 44	41. that the dolphin had been separated from its family.
4	38, 41	43, 44	42. They were able to lead it back to its family.
7	17, 19, 42	44, 45,46, 48	43. "I'm so happy
5	41, 42, 43	45, 50	44. we were able to help the dolphin,"
4	43, 44	46, 47	45. said Sue.
5	43, 45	47, 48, 49	46. "Me, too!"
3	45, 46	48	47. said Annie,
4	43, 46, 47	49	48. "And I'm happy
5	5, 13, 46, 48	50	49. we met each other, too! [GOAL ATTAINMENT]
4	5,17, 49, 44		50. This turned out to be a great day!"

HC Cutoff	HC w/ Goal statement + Goal attainment divided by story units	% HC	HC w/o Goal statement + Goal attainment divided by story units	% HC
6	13/50	26%	11/50	22%

Recall Coding Glossary for Annie

Story unit #	**Conservative (C)** thought unit from a subject transcript	Rationale
13	and she met their daughter. Um the fisherman's daughter	**She** met close enough to **they** met
16	About dolphins	
20	And ~~her friend~~ Sue said	Exact match to ***Sue said***. "Her friend" is a separate thought unit – See NMC.
21	And then Sue said that she saw a dolphin all alone	All dolphins are always swimming, so **a dolphin all alone** is essentially equivalent to a dolphin **swimming all by itself**.
30	Third, umm... She went to the library	Directly matches key words in the story unit. Still a conservative match even though some words (ideas) are left out: ***small*** library ***in town***.

Story unit #	**Liberal (L)** thought unit from a subject transcript	Rationale
4	And then she didn't have no friends (1278)	Idiomatic way of saying "she didn't have any friends," so it's liberal match.
8	Swimming	While "swimming" is a direct, literal match to the first word of the story unit, ***in groups*** is important to the story. Consequently, this is a liberal match.
12	And then once her dad um was with a fisherman	***with a fisherman*** is similar to, but not necessarily the same **as *met a fisherman***. Some interpretation is involved.
13	And then she met Sue	The fact that Sue is a daughter is critical to a conservative match. Without including daughter/ child/kid, it is a liberal match.
15	And they talked a lot	Captures the gist of ***they began to talk***.
15	And they um well um she-they were talking (1096)	Slightly different from ***began to talk*** but a match
17, 19	Um... they all three of them like to watch the dolphins swimming (1502)	Code both elements of dialogue that indicates they both like to watch dolphins
20	And then they discovered that, then Sue discovered that	In the story, ***said Sue*** was referring to Sue telling about about Sue discovering a dolphin that was swimming alone. Functionally matched to "said Sue."
27, 28	Then they wanted to research about dolphins	***Wanted to research*** is close to ***Let's see*** *("wanted")* ***if we can learn more*** *("to research"). It crosses two SUs, but is a close enough match to qualify for L rather than NMC.*
30	Then they wanted to go to the library	The difference between ***wanted to*** and ***went*** makes it a liberal match

Recall Coding Glossary for Annie

30	Go to the small town library (1096)	The student posed this as a question "maybe we should go to the small town library" in the story they **went** t o the small library in town; suggestion vs actual behavior = Liberal coding
38	Fourth, umm... they called the animal rescue office	The story said they ***decided to call*** the animal rescue office. This thought unit says ***they called***. Close enough to the gist of SU38 to be L.
40	And they had found out (1065)	The story said ***the rescuers discovered***. Slightly different from ***they found out.***
42	Then..um... they caught.. they... rescuers? And they helped the dolphin go back to his family.	Ignore the early false starts and it becomes a nice paraphrase of SU#42.
47	They said (1495)	
48	And they felt happy about it. (1495)	
50	Uh Annie and Sue um, well Annie she actually had a great day. (1124)	Infer from Annie saying that "this turned out to be a great day" that she actually had a great day. Also, the story doesn't say Sue had a great day (just that she was happy they helped the dolphin). By Annie saying she *actually* had a great day implies something additional happened, but we don't know what. Close enough for Liberal.

Story unit #	**Matched but Incorrect (MI) thought unit from a subject transcript**	**Rationale**

No Match Consistent (NMC)	**Rationale**
Two, she meet a new friend	No story unit explicitly says she made a friend or that Sue was a friend. ***Friend*** was inferred from whole story.
And they did.	Is consistent with the story from SU#30-33, but does not match any one or two SUs. No match, but consistent with (constrained by) the story.
and um they came and (1475)	Not explicitly stated in any one SU, but an accurate (constrained by) inference. Rescuers had to come to discover that the dolphin ...
~~And~~ her friend ~~Sue said~~	The story never said they were friends, but it is consistent/constrained with/by the story. The arc (scoop) separates these words from the rest of the thought unit
They both loved dolphins	Story only talks about both girls loving to watch dolphins

No Match Inconsistent (NMI)	**Rationale**